

AS level economics

chp: Demand and Supply

Topic: Income Elasticity of demand

alpha

Zarak
Mushtaq



Income Elasticity of demand (YED)

YED:- It measures the responsiveness to the percentage change in quantity demanded to the percentage change in income.

measures
responsiveness

Income and Demand
(Y)

$$YED:- \frac{\% \Delta \text{ in } QD}{\% \Delta \text{ in } Y} = +ve, -ve, 0$$

+ve:- Normal goods

-ve:- Inferior goods

0:- Necessity goods

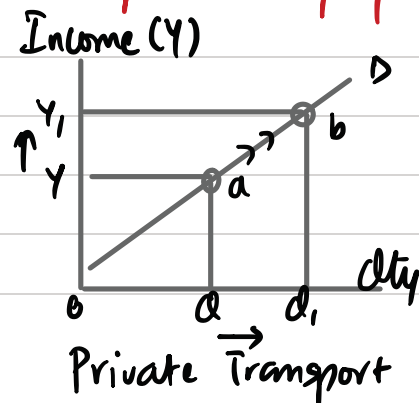
Normal goods

Income ↑
Demand ↑

+ve

Relationship

Upward sloping



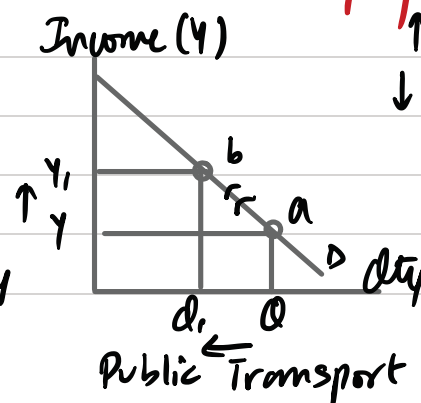
Inferior goods

Income ↑
Demand ↓

-ve

Relationship

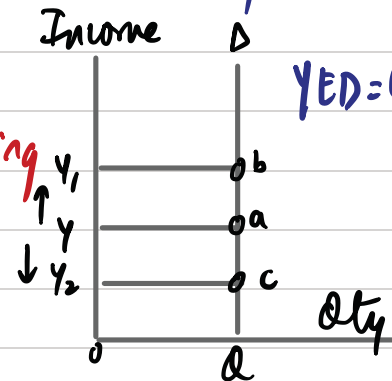
Downward sloping



Necessity goods

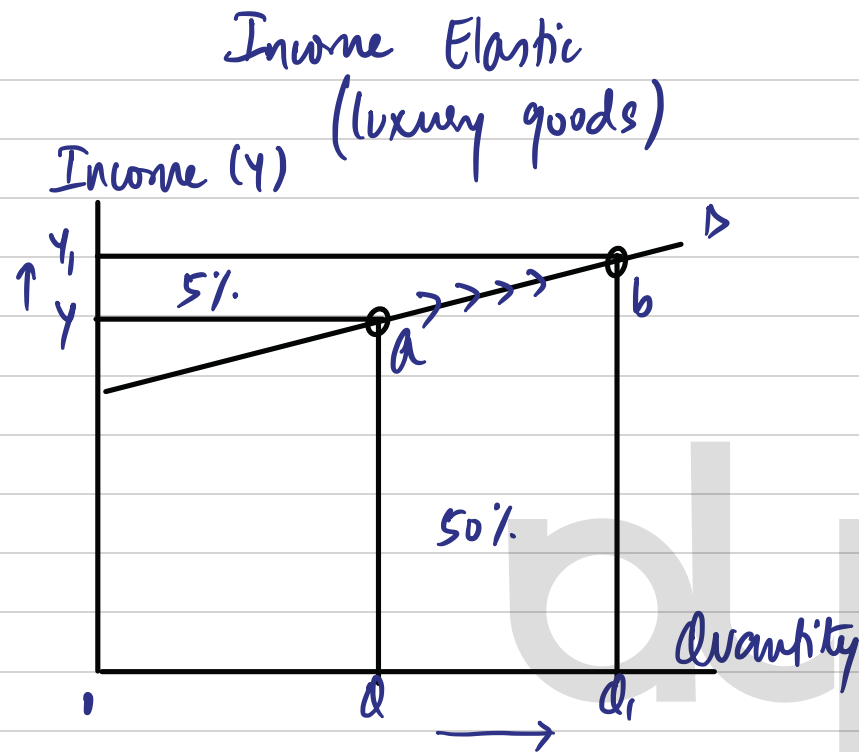
Income ↑/↓
Unchanged QD

YED=0

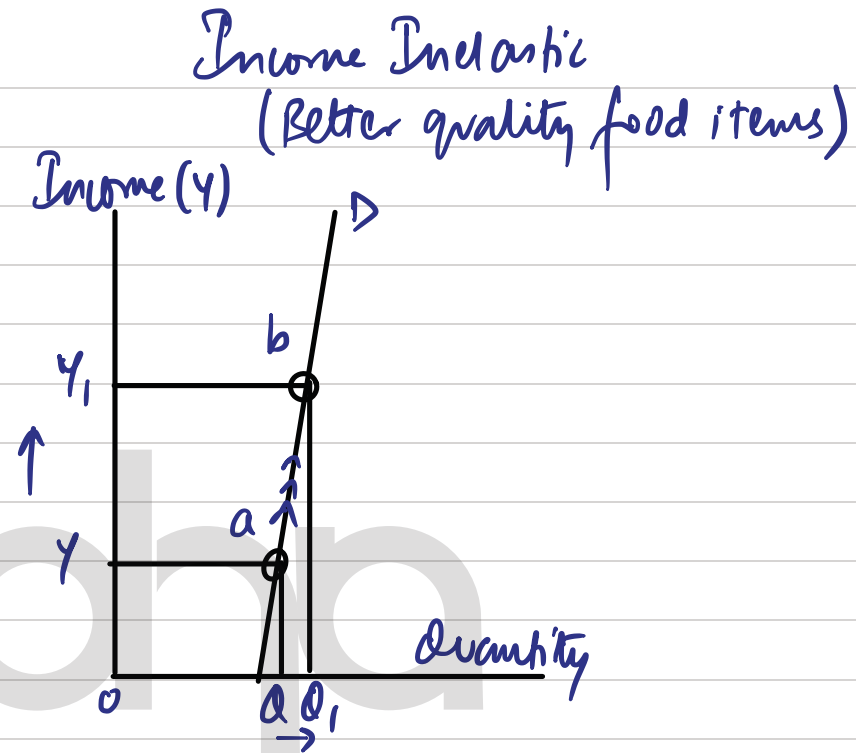


c.g. Basic
Necessity
goods

Define and
Diagram
explanation
already covered
earlier!



A small percentage change in income brings a greater percentage change in quantity demand.



A larger percentage change in income brings a smaller percentage change in quantity demand.

The factors affecting income elasticity of demand:

You need to understand that the proportion of income that is spent on a particular good or service will be likely to affect the income elasticity of demand for the product. For example, with products that are inexpensive, the income elasticity of demand will be very low, that is very inelastic.

The width of the definition of a product is likely to influence the income elasticity of demand. For example, the income elasticity of demand for cars is likely to be positive, but it may be negative for cheaper models of cars, the demand for which may fall as incomes in an economy rise.

You also need to appreciate that the economic development of a particular economy may influence the income elasticity of demand for a product. For e.g. in some economies a bicycle may be considered as a normal good, but as the economy develops and income rises, more people will be able to afford to buy a car and so the demand for bicycles may fall as incomes rise.

Usefulness of Income elasticity of demand to a producer

↓
Information

↓
Knowledge of income and
types of goods

↓
 $TR = \max$

Producer → e.g. Car showroom owner

Branded cars

2nd Hand cars

Economy is in Boom → High incomes → Income ↑ Demand ↑ $TR \uparrow$
(Producer can charge higher prices) (Branded cars)

Economy is in Recession → Low incomes → Incomes ↓ Demand ↑ $TR \uparrow$
(Producer can charge lower prices) (2nd Hand cars)

Usefulness of YED to a producer:- Economist can gain a lot of information about different types of goods based on how consumer's demand for different goods increases or decreases in response to a change in the consumer's income. For example, new car retailers may be interested in how the quantity demanded of new cars in a specific area is changing. In a case like this, we can look at consumer's income.

If the area is growing, and incomes are increasing, we can assume that more new cars will be demanded. On the other hand, if incomes are decreasing, we can anticipate that more people will buy secondhand automobiles or take public transportation. By looking at the income elasticity, we can measure the responsiveness of the quantity demanded of a good due to a change in income. We can then classify the good as normal, inferior, luxury or necessity.

Effectiveness :- Income changes can affect individuals differently
 Switching production process from normal to inferior is not easy
 Time consuming
 Expensive
 Capital required

Usefulness of Income elasticity of demand to a government

Knowledge

Information regarding
different level of income
and purchases!

objective!
Re-distribution of income
and wealth.

Higher income earners → Progressive tax → e.g. Direct tax → Income ↑
(Define) Tax % ↑
↓
Generate funds
← Known as Transfer payments (Define)
← Provided to Unemployed
← Increasing Standard of Living

Low income earners → Reduce Regressive tax → e.g. Sales tax → goods affordable
(Define)
↓
Total Expenditure ↓
Lower class will be better off

Progressive tax - A tax that takes a larger percentage of income from high income groups than from low income groups. With progressive tax, the marginal rate of tax rises as income rises i.e. as people earn more income, the rate of tax on each extra pound goes up. This causes a rise in the average rate of tax.

Proportional tax - A tax that takes the same percentage of income from all income groups. With a proportional tax, the marginal rate of tax is constant.

Regressive tax - A tax that takes a large percentage of income from low income groups than from high income groups. With regressive tax, the rate of tax falls as income rises, i.e. the average rate of tax is lower for people of higher incomes.

Progressive taxes are seen as reducing inequalities in income distribution, whereas regressive taxes can have the effect of increasing inequalities. The taxes that are generally considered progressive include income taxes and estate taxes.

A transfer payment is a payment of money for which there are no goods or services exchanged. It commonly refers to efforts by local, state and federal governments to redistribute money to those in need. In the U.S., social security, unemployment benefits, retirement pension are examples of transfer payments. Transfer payment is a redistribution of income and wealth by means of the government making a payment, without a good or services being received in return.

Questions

Marks:- 36

Q1 Using examples and diagrams, explain the values of YED? 8

Ans:- Define YED, formula, values :- +ve, -ve, 0

Define normal inferior and Necessity, e.g. + Diagrams

Q2 Discuss whether PED is more useful than YED in maximizing TR? 12

Ans:- Usefulness of PED (4) → Strategy for elastic ($P \downarrow Q \uparrow TR \uparrow$)
strategy for inelastic ($P \uparrow Q \downarrow TR \uparrow$)] Diagrams!

Usefulness of YED (4) - Paragraph given

Conclusion:- PED or YED? 4marks

Price elasticity of demand is more useful than income elasticity of demand because prices are in control of the producer where as income is beyond the control of the business. Changes in economic condition will affect individual differently making it difficult to set a product according to income. Also changing the production process from normal to inferior or other way round will be difficult, importantly research will be conducted which might be inaccurate or out-dated! Time consuming, Expensive, capital required!

Q3 Explain how information of YED will be useful for government ? 8

Define progressive, Regressive and transfer payment

State the objective

Explain the strategy to achieve the objective

Q4 A study estimates that dark chocolate has PED of -0.8 and white chocolate has -1.4 . Explain how chocolate producer can change the price to increase TR ? 8

Ans:- Define PED, formula, values, Define elastic and inelastic

Aim of the producer

-0.8 , inelastic, strategy ($P \uparrow Q \downarrow TR \uparrow$)
 -1.4 , elastic, strategy ($P \downarrow Q \uparrow TR \uparrow$)] Diagrams